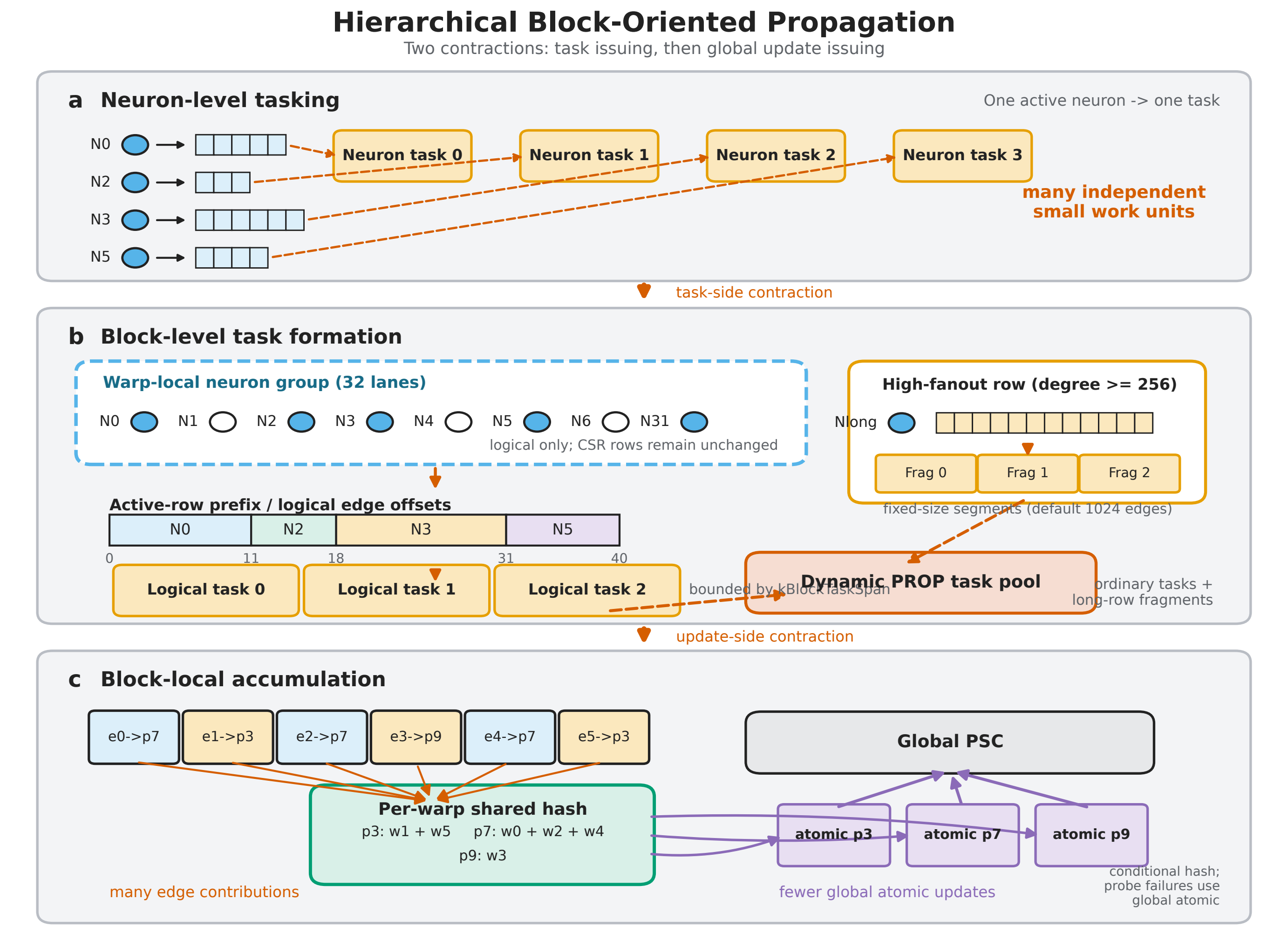
Hierarchical Block-Oriented Propagation
Two contractions: task issuing, then global update issuing
a Neuron-level tasking
One active neuron -> one task
N0
N2
N3
N5
Neuron task 0
Neuron task 1
Neuron task 2
Neuron task 3
many independent small work units
task-side contraction
b Block-level task formation
Warp-local neuron group (32 lanes)
N0 N1 N2 N3 N4 N5 N6 N31
logical only; CSR rows remain unchanged
High-fanout row (degree >= 256)
Nlong
Frag 0
Frag 1
Frag 2
fixed-size segments (default 1024 edges)
Active-row prefix / logical edge offsets
N0 N2 N3 N5
0 11 18 31 40
Logical task 0
Logical task 1
Logical task 2
bounded by kBlockTaskSpan
Dynamic PROP task pool
ordinary tasks + long-row fragments
update-side contraction
c Block-local accumulation
e0->p7
e1->p3
e2->p7
e3->p9
e4->p7
e5->p3
Per-warp shared hash
p3: w1 + w5
p7: w0 + w2 + w4
p9: w3
many edge contributions
Global PSC
atomic p3
atomic p7
atomic p9
fewer global atomic updates
conditional hash; probe failures use global atomic